

(TR-104) PYTHON WITH AI/ML

Training Day 102 Report

Date: 2 June 2026

Topic Covered: Project Presentation Preparation, AI Teaching Assistant Enhancements & System Integration Improvements

The hundred and second day of training focused on preparing the project presentation, improving the AI Teaching Assistant module, integrating RAG-based document processing, and refining the overall system workflow for better functionality and clarity.

- Prepared the project presentation covering all major modules, workflows, and system architecture.
- Organized presentation content in a structured and easy-to-understand format for effective delivery during viva.
- Reviewed and refined project concepts to ensure clarity and proper explanation in slides.
- Improved slide content to better represent technical workflows and system design.
- Implemented suggested enhancements in the project and added new feature improvements.
- Updated attendance records to store and maintain student names properly.
- Enhanced attendance logs to accurately display present and absent students.
- Integrated Retrieval-Augmented Generation (RAG) into the AI Teaching Assistant module.
- Added PDF upload functionality for processing course materials.
- Enabled document-based question answering from uploaded files.
- Unified conversational interface for handling attendance queries, database queries, and document-based queries.
- Reviewed and refined agent workflow, tool integration, and overall system architecture for better performance.
- Prepared daily diary documentation according to required submission format.

- Organized daily work entries and ensured accuracy and completeness of project records.
- Revised concepts related to RAG workflow, tool calling mechanisms, agent-based decision systems, vector database integration, and document retrieval techniques.

Overall, the training enhanced understanding of RAG-based AI systems, conversational agents, document intelligence workflows, and integrated AI application development.